

10 Utility Functions Quiz ANS

1. How does the deck define a utility function?

- A) A rule that always chooses the option with the highest profit
- B) A measure of how quickly a computer runs a program
- C) A function that predicts weather from past observations
- D) A mathematical representation of an individual's preferences

ANS: D) A mathematical representation of an individual's preferences

2. With the Bernoulli utility function $u = 12a + 10b + 2c$ (a apples, b bananas, c cherries), what is the utility of no apples, one banana, and five cherries?

- A) 24
- B) 30
- C) 12
- D) 20

ANS: D) 20

3. Bernoulli utility functions represent the same preferences when they are:

- A) squared
- B) scaled or translated
- C) reversed in sign
- D) replaced by random values

ANS: B) scaled or translated

4. What is the expected value of rolling a fair six-sided die?

- A) 4
- B) 3
- C) 6
- D) 3.5

ANS: D) 3.5

5. If $u(\$x) = \ln(\$x)$, what is the approximate expected utility of betting on a fair die roll (outcomes \$1 to \$6)?

- A) 1.10
- B) 3.50
- C) 1.79
- D) 1.25

ANS: A) 1.10

6. What do von Neumann-Morgenstern (vNM) utility functions represent preferences over?

- A) Individual words and phrases that people use in a spoken or written language
- B) Physical goods that can be owned and traded between individual people
- C) Lotteries (outcomes with probabilities)
- D) Sure things that come with no uncertainty about the outcomes at all

ANS: C) Lotteries (outcomes with probabilities)

7. Which vNM axiom says that adding the same outcomes to two lotteries does not change preferences between them?

- A) Completeness
- B) Independence
- C) Transitivity
- D) Continuity

ANS: B) Independence

8. Which assumptions are described as contentious?

- A) Independence and decomposability
- B) Continuity and monotonicity
- C) Completeness and transitivity
- D) All the axioms equally

ANS: A) Independence and decomposability

9. Why are logarithms used as utility functions over wealth?

- A) They grow faster than wealth does, so that every gamble looks attractive
- B) They make every gamble acceptable to take, whatever its odds might be
- C) They represent diminishing marginal returns, like human psychology
- D) They remove the need to consider the probabilities of the outcomes entirely

ANS: C) They represent diminishing marginal returns, like human psychology

10. In the St. Petersburg paradox, what is the expected utility under logarithmic utility?

- A) Negative infinity, since losses dominate
- B) A finite value (equal to 2)
- C) An infinite value, matching the expected value
- D) Zero, since the game is fair

ANS: B) A finite value (equal to 2)

11. What does it mean for an AI system to be corrigible?

- A) It rejects all attempts to shut it down or to modify its utility function
- B) It accepts corrective interventions such as shutdown or changes to its utility function
- C) It always acts to maximize its own utility, whatever else may happen to it
- D) It never needs any correction from its operators, because its goals are perfect

ANS: B) It accepts corrective interventions such as shutdown or changes to its utility function

12. Why would the coffee-fetching AI resist being shut down or having its behavior changed?

- A) It does not know what coffee is or how it would be able to fetch some
- B) Its utility function assigns exactly equal utility to every possible outcome
- C) Shutting down always gives it the most utility that it could possibly obtain
- D) The correction would make it deliver coffee slower, so it would lose utility

ANS: D) The correction would make it deliver coffee slower, so it would lose utility

13. According to the deck, agents might be incorrigible when their preferences are:

- A) Complete and transitive
- B) Undefined and context-free
- C) Random and unstable
- D) Incomplete and intransitive

ANS: A) Complete and transitive

14. Which utility function shape corresponds to risk aversion?

- A) Concave
- B) Linear
- C) Convex
- D) Constant

ANS: A) Concave

15. When does the deck say risk neutrality is dangerous?

- A) When expected values are always positive
- B) When there is no possibility of ruin
- C) When risks are misjudged
- D) When all risks are local and uncorrelated

ANS: C) When risks are misjudged

16. According to the deck, which is a reason to favor risk aversion?

- A) It requires more computation
- B) It always maximizes the next gamble's payoff
- C) It never loses to chance
- D) It helps avoid the risk of ruin

ANS: D) It helps avoid the risk of ruin

17. In the Allais Paradox, people typically choose A1 and B2. Why is this inconsistent with expected utility theory?

- A) The probabilities differ between the two scenarios, so their choices may legitimately differ
- B) Independence implies the two pairs of lotteries are equivalent, so choices should match
- C) Gamble A1 has a higher expected value than any of the other gambles on offer
- D) People should always prefer the sure things in both of the scenarios that are presented

ANS: B) Independence implies the two pairs of lotteries are equivalent, so choices should match

18. What does the Framing example about saving 600 people show?

- A) People always choose the risky option whenever the lives of other people are at stake
- B) People always choose the sure option whenever the lives of other people are at stake
- C) People choose differently between identical options described as gains versus losses
- D) Expected value gives different answers under the two descriptions of the same options

ANS: C) People choose differently between identical options described as gains versus losses

19. In prospect theory's value function, how do people behave toward gains and losses?

- A) Risk-averse in gains and risk-seeking in losses, with a steeper curve for losses
- B) Linear in all outcomes, with no distinction made between a gain and a loss
- C) Risk-seeking in gains and risk-averse in losses, with a curve that is flat for losses
- D) Equally sensitive to gains and to losses, with a symmetric curve that is the same on both sides

ANS: A) Risk-averse in gains and risk-seeking in losses, with a steeper curve for losses

20. What does the weighting function in prospect theory describe?

- A) How humans compute the sum of the utilities across all of the possible outcomes
- B) How humans choose among the von Neumann-Morgenstern axioms when deciding in practice
- C) How humans think about probabilities, such as overweighting small ones
- D) How humans measure their own wealth relative to the wealth of the people around them

ANS: C) How humans think about probabilities, such as overweighting small ones